

Power Series and Interval of Convergence

ANSWER KEY



Finding the radius of convergence

- 1 Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{n}$ using the Ratio Test.

$$\lim_{n \rightarrow \infty} \left| \frac{x^{n+1}/(n+1)}{x^n/n} \right| = \lim_{n \rightarrow \infty} \frac{n|x|}{n+1} = |x|. \text{ Converges when } |x| < 1: \text{ radius of convergence } R = 1.$$

- 2 Find the radius of convergence of $\sum_{n=0}^{\infty} \frac{x^n}{n!}$.

$$\lim_{n \rightarrow \infty} \left| \frac{x^{n+1}/(n+1)!}{x^n/n!} \right| = \lim_{n \rightarrow \infty} \frac{|x|}{n+1} = 0 < 1 \text{ for all } x. \text{ Radius of convergence } R = \infty \text{ (converges everywhere).}$$

Finding the interval of convergence (testing endpoints)

- 3 Find the full interval of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{n}$ (radius $R = 1$ found earlier), by testing $x = 1$ and $x = -1$.

At $x = 1$: $\sum \frac{1}{n}$ diverges (harmonic). At $x = -1$: $\sum \frac{(-1)^n}{n}$ converges (alternating harmonic, by AST). Interval of convergence: $[-1, 1)$.

- 4 Find the interval of convergence of $\sum_{n=1}^{\infty} \frac{(x-2)^n}{n^2}$.

Ratio Test: $\lim \left| \frac{(x-2)^{n+1}/(n+1)^2}{(x-2)^n/n^2} \right| = |x-2| < 1 \Rightarrow 1 < x < 3$. At $x = 1$: $\sum \frac{(-1)^n}{n^2}$ converges absolutely. At $x = 3$: $\sum \frac{1}{n^2}$ converges. Interval: $[1, 3]$.

Representing functions as power series

- 5 Write the power series for $\frac{1}{1-x}$ using the geometric series formula, and state its interval of convergence.

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + \dots, \text{ valid for } |x| < 1, \text{ i.e. interval of convergence } (-1, 1).$$

- 6 Use the geometric series for $\frac{1}{1-x}$ to find a power series for $\frac{1}{1+x^2}$.

$$\text{Substitute } x \rightarrow -x^2: \frac{1}{1-(-x^2)} = \frac{1}{1+x^2} = \sum_{n=0}^{\infty} (-x^2)^n = \sum_{n=0}^{\infty} (-1)^n x^{2n} = 1 - x^2 + x^4 - x^6 + \dots$$

Differentiating and integrating power series

- 7 Using $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$, find a power series for $\frac{1}{(1-x)^2}$ by differentiating term by term.

$$\frac{d}{dx} \left[\frac{1}{1-x} \right] = \frac{1}{(1-x)^2}. \text{ Differentiating the series: } \sum_{n=1}^{\infty} n x^{n-1}. \text{ So } \frac{1}{(1-x)^2} = \sum_{n=1}^{\infty} n x^{n-1} = 1 + 2x + 3x^2 + \dots$$

- 8 Using $\frac{1}{1+x^2} = \sum_{n=0}^{\infty} (-1)^n x^{2n}$, find a power series for $\tan^{-1} x$ by integrating term by term.

$$\tan^{-1} x = \int \frac{1}{1+x^2} dx = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1} + C. \text{ Since } \tan^{-1} 0 = 0, C = 0: \tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \dots$$

More radius/interval of convergence practice

- 9 Find the radius and interval of convergence of $\sum_{n=0}^{\infty} n! x^n$.

$\lim_{n \rightarrow \infty} \left| \frac{(n+1)! x^{n+1}}{n! x^n} \right| = \lim_{n \rightarrow \infty} (n+1)|x| = \infty$ for any $x \neq 0$. Converges only at $x = 0$: $R = 0$, interval $\{0\}$.

- 10 Find the radius and interval of convergence of $\sum_{n=1}^{\infty} \frac{(-1)^n (x+1)^n}{n \cdot 3^n}$.

$\lim_{n \rightarrow \infty} \left| \frac{(x+1)^{n+1}/[(n+1)3^{n+1}]}{(x+1)^n/(n3^n)} \right| = \frac{|x+1|}{3} < 1 \Rightarrow |x+1| < 3 \Rightarrow -4 < x < 2$; $R = 3$. At $x = 2$: $\sum \frac{(-1)^n}{n}$ converges. At $x = -4$: $\sum \frac{1}{n}$ diverges. Interval: $(-4, 2]$.

Visualizing a power series approximation

- 11 The geometric series $\frac{1}{1-x} = 1 + x + x^2 + x^3 + \dots$ converges on $(-1, 1)$. Compare the graph of $f(x) = \frac{1}{1-x}$ to its degree-3 partial sum $P_3(x) = 1 + x + x^2 + x^3$ on $[-0.8, 0.8]$, and describe where they agree most closely.

The two curves nearly overlap near $x = 0$ and diverge visibly as $x \rightarrow \pm 0.8$ (closer to the boundary of the interval of convergence), since the partial sum's approximation quality worsens near the edges of convergence.

Term-by-term operations preserve the radius of convergence

- 12 State the theorem describing what happens to the radius of convergence when a power series is differentiated or integrated term by term.

Differentiating or integrating a power series term by term does not change its radius of convergence R (though convergence AT the endpoints $x = \pm R$ may change).

- 13 Given that $\sum_{n=1}^{\infty} \frac{x^n}{n}$ has radius of convergence $R = 1$, state the radius of convergence of its derivative $\sum_{n=1}^{\infty} x^{n-1}$ without recomputing.

By the term-by-term differentiation theorem, the radius of convergence is unchanged: $R = 1$.

Building a new series by combining known ones

- 14 Using $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$ and $\frac{1}{1+x} = \sum_{n=0}^{\infty} (-1)^n x^n$, find the power series for $\frac{2}{1-x^2}$ by adding the two series.

$\frac{1}{1-x} + \frac{1}{1+x} = \frac{(1+x) + (1-x)}{1-x^2} = \frac{2}{1-x^2}$. Adding the series: $\sum_{n=0}^{\infty} x^n + \sum_{n=0}^{\infty} (-1)^n x^n = \sum_{n=0}^{\infty} [1 + (-1)^n] x^n$, which is $2x^n$ for even n and 0 for odd n : $2 + 2x^2 + 2x^4 + \dots$.

Ratio Test with factorial and power combinations

- 15 Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{2^n x^n}{n^2}$.

$$\lim_{n \rightarrow \infty} \left| \frac{2^{n+1} x^{n+1} / (n+1)^2}{2^n x^n / n^2} \right| = \lim_{n \rightarrow \infty} \frac{2n^2 |x|}{(n+1)^2} = 2|x| < 1 \Rightarrow |x| < \frac{1}{2}. R = \frac{1}{2}.$$

- 16 For the series above, test both endpoints $x = \frac{1}{2}$ and $x = -\frac{1}{2}$ to find the full interval of convergence.

At $x = \frac{1}{2}$: $\sum \frac{2^n (1/2)^n}{n^2} = \sum \frac{1}{n^2}$, converges (p-series, $p = 2$). At $x = -\frac{1}{2}$: $\sum \frac{(-1)^n}{n^2}$, converges absolutely. Interval: $[-\frac{1}{2}, \frac{1}{2}]$.

Using a power series to approximate a definite integral

- 17 Using the power series $\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \dots$ found earlier, estimate $\tan^{-1}(0.5)$ using the first three terms.

$$\tan^{-1}(0.5) \approx 0.5 - \frac{(0.5)^3}{3} + \frac{(0.5)^5}{5} = 0.5 - 0.04167 + 0.00625 \approx 0.4646 \text{ (true value } \approx 0.4636, \text{ a close match)}.$$

Multiple choice — easy 15

- 18 Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{6^n}$.

a. 12

b. $\frac{1}{6}$

c. 6 ✓

d. 1

$$\lim_{n \rightarrow \infty} \left| \frac{x^{n+1}/6^{n+1}}{x^n/6^n} \right| = \frac{|x|}{6} < 1 \Rightarrow |x| < 6. R = 6.$$

- 19 Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{9^n}$.

a. 9 ✓

b. $\frac{1}{9}$

c. 18

d. 1

$$\lim_{n \rightarrow \infty} \left| \frac{x^{n+1}/9^{n+1}}{x^n/9^n} \right| = \frac{|x|}{9} < 1 \Rightarrow |x| < 9. R = 9.$$

20 Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{18^n}$.

a. 1

b. 18 ✓

c. 36

d. $\frac{1}{18}$

$$\lim_{n \rightarrow \infty} \left| \frac{x^{n+1}/18^{n+1}}{x^n/18^n} \right| = \frac{|x|}{18} < 1 \Rightarrow |x| < 18. R = 18.$$

21 Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{13^n}$.

a. $\frac{1}{13}$

b. 26

c. 1

d. 13 ✓

$$\lim_{n \rightarrow \infty} \left| \frac{x^{n+1}/13^{n+1}}{x^n/13^n} \right| = \frac{|x|}{13} < 1 \Rightarrow |x| < 13. R = 13.$$

22 Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{5^n}$.

a. 1

b. 10

c. 5 ✓

d. $\frac{1}{5}$

$$\lim_{n \rightarrow \infty} \left| \frac{x^{n+1}/5^{n+1}}{x^n/5^n} \right| = \frac{|x|}{5} < 1 \Rightarrow |x| < 5. R = 5.$$

23 Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{7^n}$.

a. $\frac{1}{7}$

b. 1

c. 14

d. 7 ✓

$$\lim_{n \rightarrow \infty} \left| \frac{x^{n+1}/7^{n+1}}{x^n/7^n} \right| = \frac{|x|}{7} < 1 \Rightarrow |x| < 7. R = 7.$$

24 Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{20^n}$.

a. 1

b. 20 ✓

c. 40

d. $\frac{1}{20}$

$$\lim_{n \rightarrow \infty} \left| \frac{x^{n+1}/20^{n+1}}{x^n/20^n} \right| = \frac{|x|}{20} < 1 \Rightarrow |x| < 20. R = 20.$$

25 Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{8^n}$.

a. $\frac{1}{8}$

b. 8 ✓

c. 16

d. 1

$$\lim_{n \rightarrow \infty} \left| \frac{x^{n+1}/8^{n+1}}{x^n/8^n} \right| = \frac{|x|}{8} < 1 \Rightarrow |x| < 8. R = 8.$$

26 Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{19^n}$.

a. $\frac{1}{19}$

b. 38

c. 19 ✓

d. 1

$$\lim_{n \rightarrow \infty} \left| \frac{x^{n+1}/19^{n+1}}{x^n/19^n} \right| = \frac{|x|}{19} < 1 \Rightarrow |x| < 19. R = 19.$$

27 Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{14^n}$.

a. 28

b. 14 ✓

c. $\frac{1}{14}$

d. 1

$$\lim_{n \rightarrow \infty} \left| \frac{x^{n+1}/14^{n+1}}{x^n/14^n} \right| = \frac{|x|}{14} < 1 \Rightarrow |x| < 14. R = 14.$$

28 Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{2^n}$.

a. 1

b. 4

c. $\frac{1}{2}$

d. 2 ✓

$$\lim_{n \rightarrow \infty} \left| \frac{x^{n+1}/2^{n+1}}{x^n/2^n} \right| = \frac{|x|}{2} < 1 \Rightarrow |x| < 2. R = 2.$$

29 Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{4^n}$.

a. 4 ✓

b. $\frac{1}{4}$

c. 1

d. 8

$$\lim_{n \rightarrow \infty} \left| \frac{x^{n+1}/4^{n+1}}{x^n/4^n} \right| = \frac{|x|}{4} < 1 \Rightarrow |x| < 4. R = 4.$$

30 Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{12^n}$.

a. 1

b. 24

c. $\frac{1}{12}$

d. 12 ✓

$$\lim_{n \rightarrow \infty} \left| \frac{x^{n+1}/12^{n+1}}{x^n/12^n} \right| = \frac{|x|}{12} < 1 \Rightarrow |x| < 12. R = 12.$$

31 Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{15^n}$.

a. 1

b. $\frac{1}{15}$

c. 15 ✓

d. 30

$$\lim_{n \rightarrow \infty} \left| \frac{x^{n+1}/15^{n+1}}{x^n/15^n} \right| = \frac{|x|}{15} < 1 \Rightarrow |x| < 15. R = 15.$$

32 Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{3^n}$.

a. $\frac{1}{3}$

b. 3 ✓

c. 6

d. 1

$$\lim_{n \rightarrow \infty} \left| \frac{x^{n+1}/3^{n+1}}{x^n/3^n} \right| = \frac{|x|}{3} < 1 \Rightarrow |x| < 3. R = 3.$$

Multiple choice — medium 15

33 Find the radius of convergence of $\sum_{n=1}^{\infty} 7^n(x+6)^n$.

a. $\frac{1}{7}$ ✓

b. 7

c. $\frac{8}{7}$

d. $\frac{1}{8}$

$$\lim_{n \rightarrow \infty} \left| \frac{7^{n+1}(x+6)^{n+1}}{7^n(x+6)^n} \right| = 7|x+6| < 1 \Rightarrow |x+6| < \frac{1}{7}. R = \frac{1}{7}.$$

34 Find the radius of convergence of $\sum_{n=1}^{\infty} 7^n(x-4)^n$.

a. $\frac{1}{8}$

b. $\frac{1}{7}$ ✓

c. $\frac{8}{7}$

d. 7

$$\lim_{n \rightarrow \infty} \left| \frac{7^{n+1}(x-4)^{n+1}}{7^n(x-4)^n} \right| = 7|x-4| < 1 \Rightarrow |x-4| < \frac{1}{7}. R = \frac{1}{7}.$$

35 Find the radius of convergence of $\sum_{n=1}^{\infty} 3^n(x+9)^n$.

a. $\frac{4}{3}$

b. $\frac{1}{4}$

c. $\frac{1}{3}$ ✓

d. 3

$$\lim_{n \rightarrow \infty} \left| \frac{3^{n+1}(x+9)^{n+1}}{3^n(x+9)^n} \right| = 3|x+9| < 1 \Rightarrow |x+9| < \frac{1}{3}. R = \frac{1}{3}.$$

36 Find the radius of convergence of $\sum_{n=1}^{\infty} 7^n(x+3)^n$.

a. 7

b. $\frac{8}{7}$

c. $\frac{1}{7}$ ✓

d. $\frac{1}{8}$

$$\lim_{n \rightarrow \infty} \left| \frac{7^{n+1}(x+3)^{n+1}}{7^n(x+3)^n} \right| = 7|x+3| < 1 \Rightarrow |x+3| < \frac{1}{7}. R = \frac{1}{7}.$$

37 Find the radius of convergence of $\sum_{n=1}^{\infty} 5^n(x+1)^n$.

a. $\frac{1}{5}$ ✓

b. 5

c. $\frac{6}{5}$

d. $\frac{1}{6}$

$$\lim_{n \rightarrow \infty} \left| \frac{5^{n+1}(x+1)^{n+1}}{5^n(x+1)^n} \right| = 5|x+1| < 1 \Rightarrow |x+1| < \frac{1}{5}. R = \frac{1}{5}.$$

38 Find the radius of convergence of $\sum_{n=1}^{\infty} 2^n(x+7)^n$.

a. $\frac{1}{3}$

b. $\frac{3}{2}$

c. $\frac{1}{2}$ ✓

d. 2

$$\lim_{n \rightarrow \infty} \left| \frac{2^{n+1}(x+7)^{n+1}}{2^n(x+7)^n} \right| = 2|x+7| < 1 \Rightarrow |x+7| < \frac{1}{2}. R = \frac{1}{2}.$$

39 Find the radius of convergence of $\sum_{n=1}^{\infty} 9^n(x-4)^n$.

a. $\frac{1}{10}$

b. $\frac{10}{9}$

c. $\frac{1}{9}$ ✓

d. 9

$$\lim_{n \rightarrow \infty} \left| \frac{9^{n+1}(x-4)^{n+1}}{9^n(x-4)^n} \right| = 9|x-4| < 1 \Rightarrow |x-4| < \frac{1}{9}. R = \frac{1}{9}.$$

40 Find the radius of convergence of $\sum_{n=1}^{\infty} 2^n(x+4)^n$.

a. $\frac{1}{2}$ ✓

b. 2

c. $\frac{3}{2}$

d. $\frac{1}{3}$

$$\lim_{n \rightarrow \infty} \left| \frac{2^{n+1}(x+4)^{n+1}}{2^n(x+4)^n} \right| = 2|x+4| < 1 \Rightarrow |x+4| < \frac{1}{2}. R = \frac{1}{2}.$$

41 Find the radius of convergence of $\sum_{n=1}^{\infty} 9^n(x-1)^n$.

a. $\frac{1}{9}$ ✓

b. 9

c. $\frac{10}{9}$

d. $\frac{1}{10}$

$$\lim_{n \rightarrow \infty} \left| \frac{9^{n+1}(x-1)^{n+1}}{9^n(x-1)^n} \right| = 9|x-1| < 1 \Rightarrow |x-1| < \frac{1}{9}. R = \frac{1}{9}.$$

42 Find the radius of convergence of $\sum_{n=1}^{\infty} 2^n(x+5)^n$.

a. 2

b. $\frac{3}{2}$

c. $\frac{1}{2}$ ✓

d. $\frac{1}{3}$

$$\lim_{n \rightarrow \infty} \left| \frac{2^{n+1}(x+5)^{n+1}}{2^n(x+5)^n} \right| = 2|x+5| < 1 \Rightarrow |x+5| < \frac{1}{2}. R = \frac{1}{2}.$$

43 Find the radius of convergence of $\sum_{n=1}^{\infty} 2^n(x+8)^n$.

a. $\frac{1}{2}$ ✓

b. $\frac{3}{2}$

c. $\frac{1}{3}$

d. 2

$$\lim_{n \rightarrow \infty} \left| \frac{2^{n+1}(x+8)^{n+1}}{2^n(x+8)^n} \right| = 2|x+8| < 1 \Rightarrow |x+8| < \frac{1}{2}. R = \frac{1}{2}.$$

44 Find the radius of convergence of $\sum_{n=1}^{\infty} 3^n(x+6)^n$.

a. $\frac{1}{4}$

b. 3

c. $\frac{4}{3}$

d. $\frac{1}{3}$ ✓

$$\lim_{n \rightarrow \infty} \left| \frac{3^{n+1}(x+6)^{n+1}}{3^n(x+6)^n} \right| = 3|x+6| < 1 \Rightarrow |x+6| < \frac{1}{3}. R = \frac{1}{3}.$$

45 Find the radius of convergence of $\sum_{n=1}^{\infty} 7^n(x+8)^n$.

a. $\frac{8}{7}$

b. $\frac{1}{7}$ ✓

c. 7

d. $\frac{1}{8}$

$$\lim_{n \rightarrow \infty} \left| \frac{7^{n+1}(x+8)^{n+1}}{7^n(x+8)^n} \right| = 7|x+8| < 1 \Rightarrow |x+8| < \frac{1}{7}. R = \frac{1}{7}.$$

46 Find the radius of convergence of $\sum_{n=1}^{\infty} 8^n(x-5)^n$.

a. $\frac{9}{8}$

b. $\frac{1}{8}$ ✓

c. $\frac{1}{9}$

d. 8

$$\lim_{n \rightarrow \infty} \left| \frac{8^{n+1}(x-5)^{n+1}}{8^n(x-5)^n} \right| = 8|x-5| < 1 \Rightarrow |x-5| < \frac{1}{8}. R = \frac{1}{8}.$$

47 Find the radius of convergence of $\sum_{n=1}^{\infty} 6^n(x+3)^n$.

a. $\frac{1}{6}$ ✓

b. 6

c. $\frac{1}{7}$

d. $\frac{7}{6}$

$$\lim_{n \rightarrow \infty} \left| \frac{6^{n+1}(x+3)^{n+1}}{6^n(x+3)^n} \right| = 6|x+3| < 1 \Rightarrow |x+3| < \frac{1}{6}. R = \frac{1}{6}.$$

Multiple choice — hard 15

48 Find the full interval of convergence of $\sum_{n=1}^{\infty} \frac{(x+7)^n}{3^n}$.

a. $[-10, -4]$

b. $[-10, -4)$

c. **$(-10, -4)$ ✓**

d. $(-10, -4]$

Ratio Test gives $|x + 7| < 3$, so $-10 < x < -4$ ($R = 3$). Testing endpoints: the terms don't approach 0 at either endpoint, so both diverge (by the n th-term test). Interval: $(-10, -4)$.

49 Find the full interval of convergence of $\sum_{n=1}^{\infty} \frac{(x-1)^n}{4^n n}$.

a. **$[-3, 5)$ ✓**

b. $(-3, 5]$

c. $(-3, 5)$

d. $[-3, 5]$

Ratio Test gives $|x - 1| < 4$, so $-3 < x < 5$ ($R = 4$). Testing endpoints: at $x = 5$: $\sum \frac{1}{n}$ diverges; at $x = -3$: $\sum \frac{(-1)^n}{n}$ converges by the AST. Interval: $[-3, 5)$.

50 Find the full interval of convergence of $\sum_{n=1}^{\infty} \frac{(x-8)^n}{6^n}$.

a. **$(2, 14)$ ✓**

b. $[2, 14]$

c. $[2, 14)$

d. $(2, 14]$

Ratio Test gives $|x - 8| < 6$, so $2 < x < 14$ ($R = 6$). Testing endpoints: the terms don't approach 0 at either endpoint, so both diverge (by the n th-term test). Interval: $(2, 14)$.

51 Find the full interval of convergence of $\sum_{n=1}^{\infty} \frac{(x-3)^n}{6^n n}$.

- a. $(-3,9]$
- b. $(-3,9)$
- c. $[-3,9]$ ✓
- d. $[-3,9]$

Ratio Test gives $|x - 3| < 6$, so $-3 < x < 9$ ($R = 6$). Testing endpoints: at $x = 9$: $\sum \frac{1}{n}$ diverges; at $x = -3$: $\sum \frac{(-1)^n}{n}$ converges by the AST. Interval: $[-3, 9)$.

52 Find the full interval of convergence of $\sum_{n=1}^{\infty} \frac{(x-5)^n}{4^n n}$.

- a. $(1,9]$
- b. $[1,9]$ ✓
- c. $[1,9]$
- d. $(1,9)$

Ratio Test gives $|x - 5| < 4$, so $1 < x < 9$ ($R = 4$). Testing endpoints: at $x = 9$: $\sum \frac{1}{n}$ diverges; at $x = 1$: $\sum \frac{(-1)^n}{n}$ converges by the AST. Interval: $[1, 9)$.

53 Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{n^2 \cdot 8^n x^n}{12^n}$.

- a. $\frac{5}{2}$
- b. 96
- c. $\frac{2}{3}$
- d. $\frac{3}{2}$ ✓

The polynomial factor n^2 does not affect the radius. $\lim_{n \rightarrow \infty} \left| \frac{8^{n+1} x^{n+1} / 12^{n+1}}{8^n x^n / 12^n} \right| = \frac{8}{12} |x| < 1 \Rightarrow |x| < \frac{12}{8}$. $R = \frac{3}{2}$.

54 Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{n^2 \cdot 8^n x^n}{16^n}$.

- a. 2 ✓
- b. 3
- c. $\frac{1}{2}$
- d. 128

The polynomial factor n^2 does not affect the radius. $\lim_{n \rightarrow \infty} \left| \frac{8^{n+1} x^{n+1} / 16^{n+1}}{8^n x^n / 16^n} \right| = \frac{8}{16} |x| < 1 \Rightarrow |x| < \frac{16}{8}. R = 2.$

55 Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{n^2 \cdot 5^n x^n}{7^n}$.

- a. $\frac{5}{7}$
- b. 35
- c. $\frac{12}{5}$
- d. $\frac{7}{5}$ ✓

The polynomial factor n^2 does not affect the radius. $\lim_{n \rightarrow \infty} \left| \frac{5^{n+1} x^{n+1} / 7^{n+1}}{5^n x^n / 7^n} \right| = \frac{5}{7} |x| < 1 \Rightarrow |x| < \frac{7}{5}. R = \frac{7}{5}.$

56 Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{n^2 \cdot 6^n x^n}{11^n}$.

- a. $\frac{6}{11}$
- b. $\frac{17}{6}$
- c. $\frac{11}{6}$ ✓
- d. 66

The polynomial factor n^2 does not affect the radius. $\lim_{n \rightarrow \infty} \left| \frac{6^{n+1} x^{n+1} / 11^{n+1}}{6^n x^n / 11^n} \right| = \frac{6}{11} |x| < 1 \Rightarrow |x| < \frac{11}{6}. R = \frac{11}{6}.$

57 Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{n^2 \cdot 9^n x^n}{15^n}$.

- a. $\frac{8}{3}$
- b. 135
- c. $\frac{3}{5}$
- d. $\frac{5}{3}$ ✓

The polynomial factor n^2 does not affect the radius. $\lim_{n \rightarrow \infty} \left| \frac{9^{n+1} x^{n+1} / 15^{n+1}}{9^n x^n / 15^n} \right| = \frac{9}{15} |x| < 1 \Rightarrow |x| < \frac{15}{9}. R = \frac{5}{3}.$

58 Using $\frac{1}{1-3x} = \sum_{n=0}^{\infty} 3^n x^n$, find the coefficient of x^3 .

a. 27 ✓

b. 26

c. 81

d. 9

The coefficient of x^n is 3^n , so the coefficient of x^3 is $3^3 = 27$.

59 Using $\frac{1}{1-3x} = \sum_{n=0}^{\infty} 3^n x^n$, find the coefficient of x^5 .

a. 81

b. 729

c. 15

d. 243 ✓

The coefficient of x^n is 3^n , so the coefficient of x^5 is $3^5 = 243$.

60 Using $\frac{1}{1-4x} = \sum_{n=0}^{\infty} 4^n x^n$, find the coefficient of x^7 .

a. 28

b. 16384 ✓

c. 4096

d. 65536

The coefficient of x^n is 4^n , so the coefficient of x^7 is $4^7 = 16384$.

61 Using $\frac{1}{1-5x} = \sum_{n=0}^{\infty} 5^n x^n$, find the coefficient of x^3 .

a. 25

b. 125 ✓

c. 15

d. 625

The coefficient of x^n is 5^n , so the coefficient of x^3 is $5^3 = 125$.

62 Using $\frac{1}{1-5x} = \sum_{n=0}^{\infty} 5^n x^n$, find the coefficient of x^7 .

a. 78125 ✓

b. 35

c. 390625

d. 15625

The coefficient of x^n is 5^n , so the coefficient of x^7 is $5^7 = 78125$.